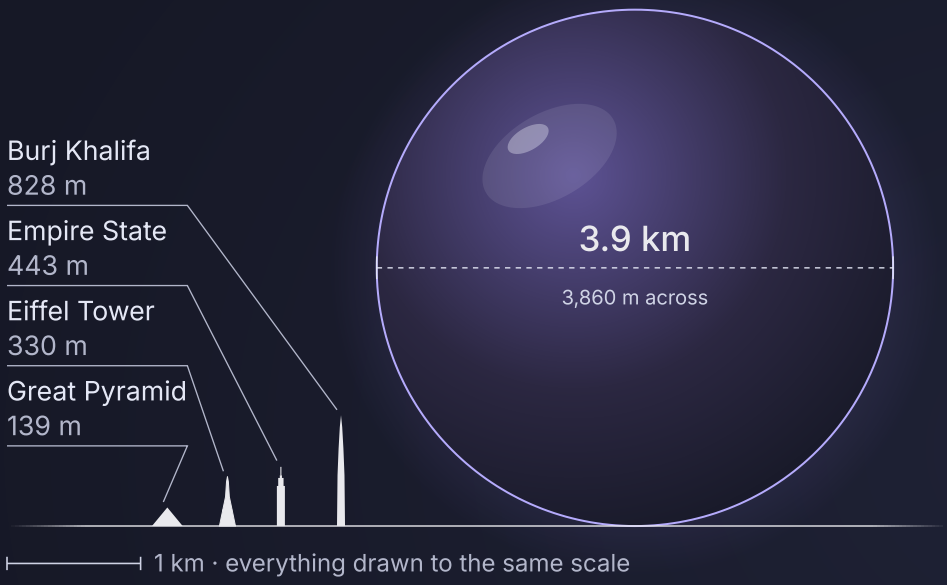
KANSAS · SOIL MOISTURE

KANSAS STATE UNIVERSITY | Agronomy
Soil Water Processes Lab

30 km³

The most plant available water the top 60 cm of Kansas soils can hold, with the whole state at field capacity.

= 30 billion m³ · = 30 trillion liters



Gathered into one marble, it would stand **more than 4½ times taller** than the world's tallest building.

12 million

Olympic swimming pools — one pool = 2,500 m³

94%

of full Lake Mead, the largest man-made reservoir in the U.S.

141 mm

of a single rainfall event soaking the entire state at once

CLOSER TO HOME

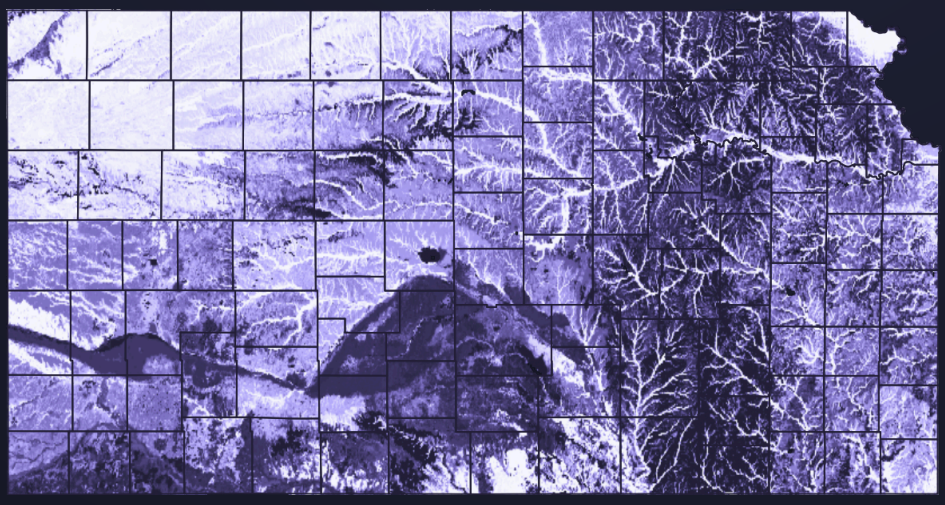
5½ times

That's how many times this water could fill every lake and pond in Kansas combined (≈ 5.5 km³).

Or, the ten largest lakes one at a time:

Milford	0.45 km³	67×
Wilson	0.29 km³	103×
Tuttle Creek	0.29 km³	106×
Waconda	0.26 km³	117×
Perry	0.24 km³	127×
Cedar Bluff	0.21 km³	145×
Cheney	0.19 km³	157×
El Dorado	0.19 km³	157×
Melvern	0.18 km³	166×
Clinton	0.14 km³	211×

WHERE IT IS STORED



≤ 125 mm ≥ 157 mm

Silt loams in the northwest hold the most plant available water; heavy clay soils in the east and sandy soils along the Arkansas River hold the least.

WHY IT MATTERS

Crops, pastures, and native grasslands depend on this stored soil water to sustain transpiration and produce grain and forage between rains. How full the soil profile is at any moment, known as the fraction of available water, is a robust indicator of agricultural drought.

Plant available water is what a soil holds after excess water drains away (field capacity) minus what it holds too tightly for roots to extract (permanent wilting point). We report the top 60 cm, the most dynamic and densely rooted part of the soil profile, where water content increases within hours of rainfall and declines over the following days through soil evaporation and plant uptake. Both limits were estimated with custom pedotransfer functions developed by the K-State Soil Water Processes Lab, using clay, sand, bulk density, and organic matter maps from the USDA Natural Resources Conservation Service. Figures rounded: full Lake Mead ≈ 32 km³; Kansas area ≈ 213,000 km²; lakes and ponds at normal pool (Rahmani et al. 2018; USACE National Inventory of Dams; Callihan 2013). Created by Andres Patrignani, Soil Water Processes Lab, Agronomy, Kansas State University, October 2026.